

Progress Test 3.B — Elec-S1
 3.B · Thévenin’s and Norton’s equivalence theorems
 Semester 1 · Academic Year 2026–2027

Family name: Given name:

Duration: 10 minutes. *No documents. Calculator not allowed.* Answer inside the printed frames only; the margin boxes are for the marker. **Total: 14 marks.**

Exercise 1. From Norton to Thévenin (1 mark)
 A Norton model has $I_{No} = 2.0\text{ mA}$ and $R_{No} = 4.7\text{ k}\Omega$. Give the electromotive force E_{Th} of the equivalent Thévenin model.

[Empty grid for Exercise 1 answer]

Marking

①	
0	1
☐	☐

Exercise 2. Thévenin’s theorem (5 marks)
 State Thévenin’s theorem for a network made of ideal sources and ohmic conductors, seen from its terminals A and B (draw the model). Explain how E_{Th} and R_{Th} are determined.

[Empty grid for Exercise 2 answer]

Marking

①		
0	1	2
☐	☐	☐

②		
0	1	2
☐	☐	☐

③	
0	1
☐	☐

Exercise 3. Thévenin and Norton models

(3 marks)

To switch off an ideal voltage source, it is replaced by:

- ☐ an open circuit
 - ☐ a short circuit (a wire)
 - ☐ a resistor R_{Th}
 - ☐ an ideal current source

A Thévenin model has $E_{\text{Th}} = 12 \text{ V}$ and $R_{\text{Th}} = 3.0 \Omega$. The equivalent Norton model has:

- ☐ $I_{No} = 36 \text{ A}, R_{No} = 3.0 \Omega$

☐ $I_{No} = 4.0 \text{ A}, R_{No} = 3.0 \Omega$

☐ $I_{No} = 4.0 \text{ A}, R_{No} = 0.33 \Omega$

☐ $I_{No} = 0.25 \text{ A}, R_{No} = 3.0 \Omega$

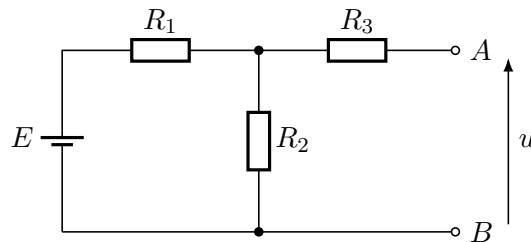
A Thévenin (or Norton) model can describe a “black box” containing:

- ☐ a diode;
- ☐ only ideal sources and ohmic conductors;
- ☐ a lamp;
- ☐ a motor.

Exercise 4. Thévenin model of a network

(5 marks)

Determine the Thévenin model ($E_{\text{Th}}, R_{\text{Th}}$) of the network below, seen from the terminals A and B . Deduce the current I_{No} of the equivalent Norton model.



$E = 12\text{ V}$	$R_1 = 6.0\text{ k}\Omega$	$R_2 = 3.0\text{ k}\Omega$	$R_3 = 1.0\text{ k}\Omega$
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[illegible]

Marking

①	
0	1

②	
0	1

③	
0	1

④	
0	1

⑤	
0	1

— End of the test —